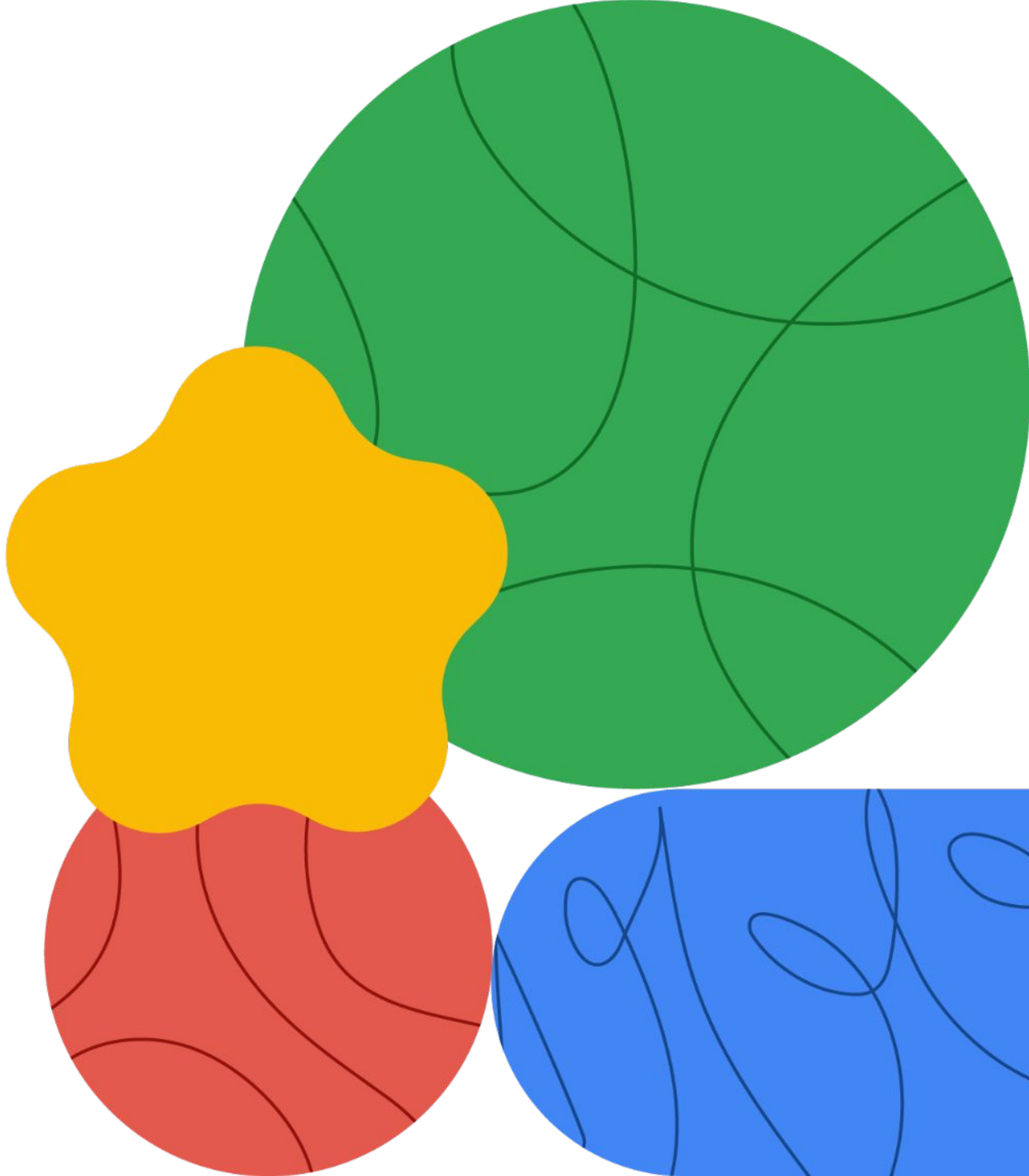
Google Cloud

Introduction to AI and Machine Learning on Google Cloud

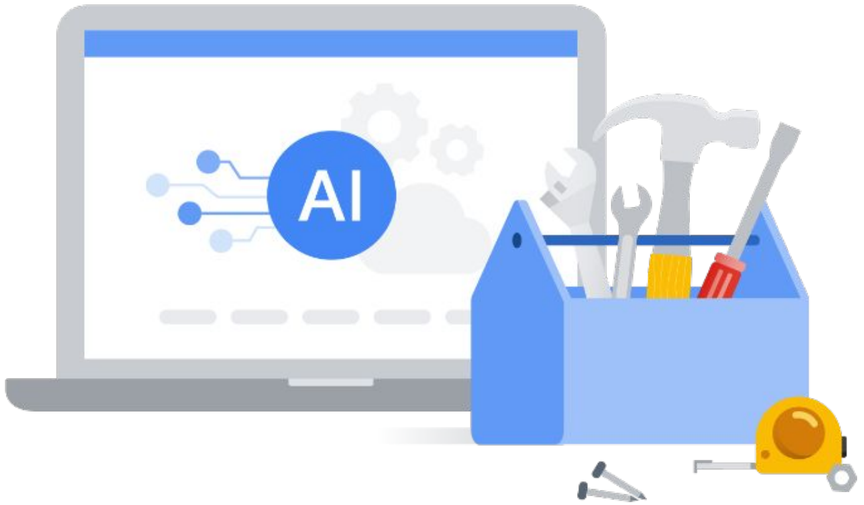
Module 3

AI development options



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AI development options



No-code	No-to-low code	Low-code	Code-based
Gemini Enterprise app Gemini Enterprise for Customer Experience	Agent Platform AutoML	Pretrained APIs	Custom training BigQuery ML ADK

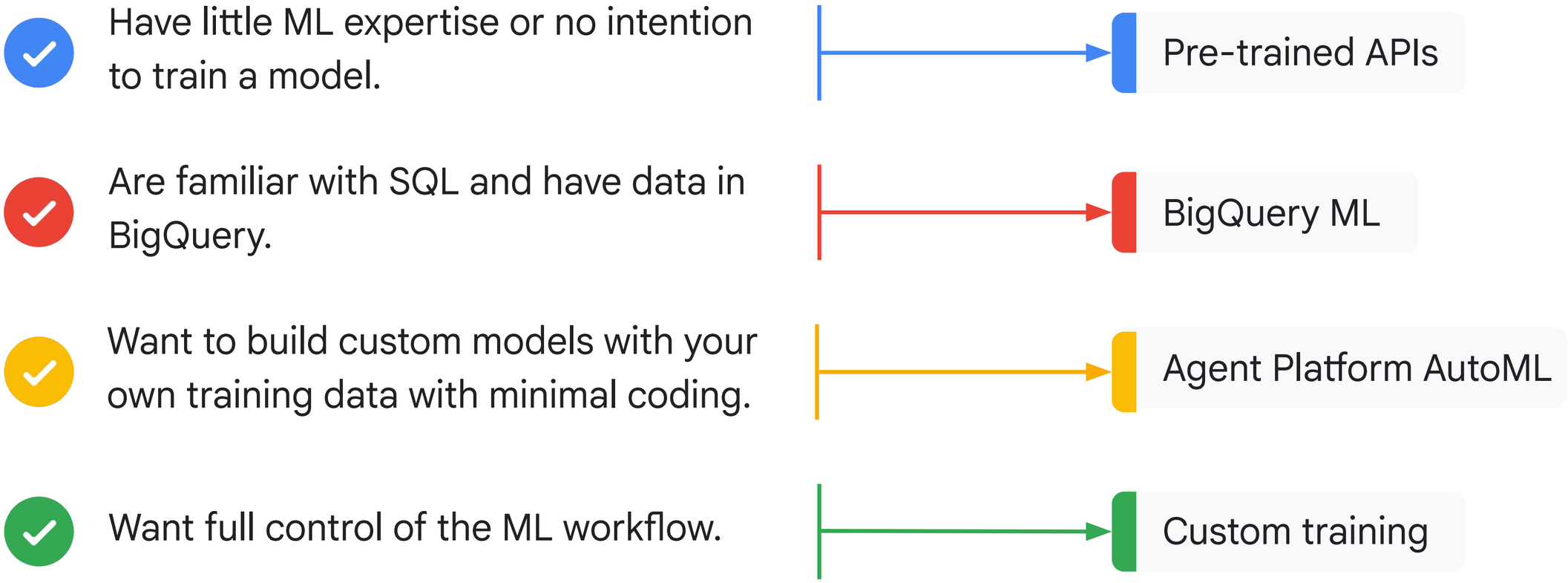
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Compare the options

	Pre-trained APIs	BigQuery ML	Agent Platform AutoML	Custom training
Data type	Tabular, image, text, audio, and video	Tabular	Tabular, image, and video	Tabular, image, text, and video
Training data size	No data required	Medium to large	Small to medium	Medium to large
ML and coding expertise	Low	Medium	Low	High
Flexibility to tune hyperparameters	None	Medium	None	High
Time to train a model	None	Medium	Medium	Long

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Choose the right option



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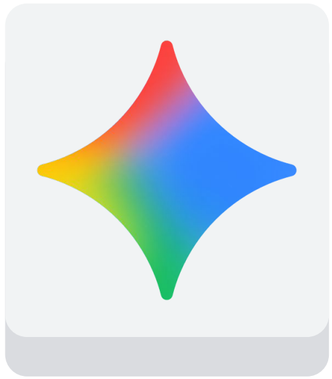


Agent Platform

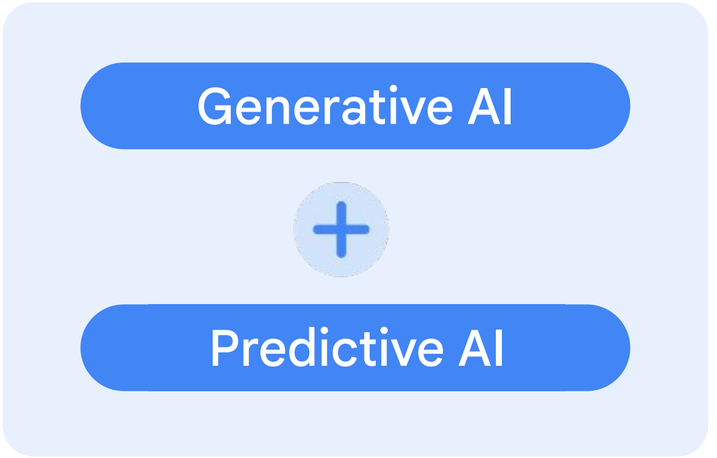
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Agent Platform: a unified AI development platform

Support both generative AI and predictive AI



Agent Platform



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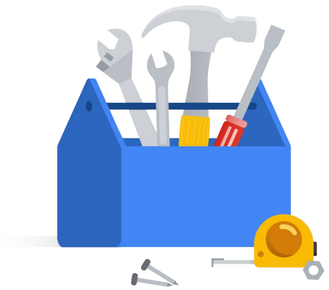
Agent Platform AutoML

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Agent Platform AutoML: Automate the ML pipeline

Data

Automated feature engineering



Automated architecture search



Automated hyperparameter tuning



Automated model ensemble



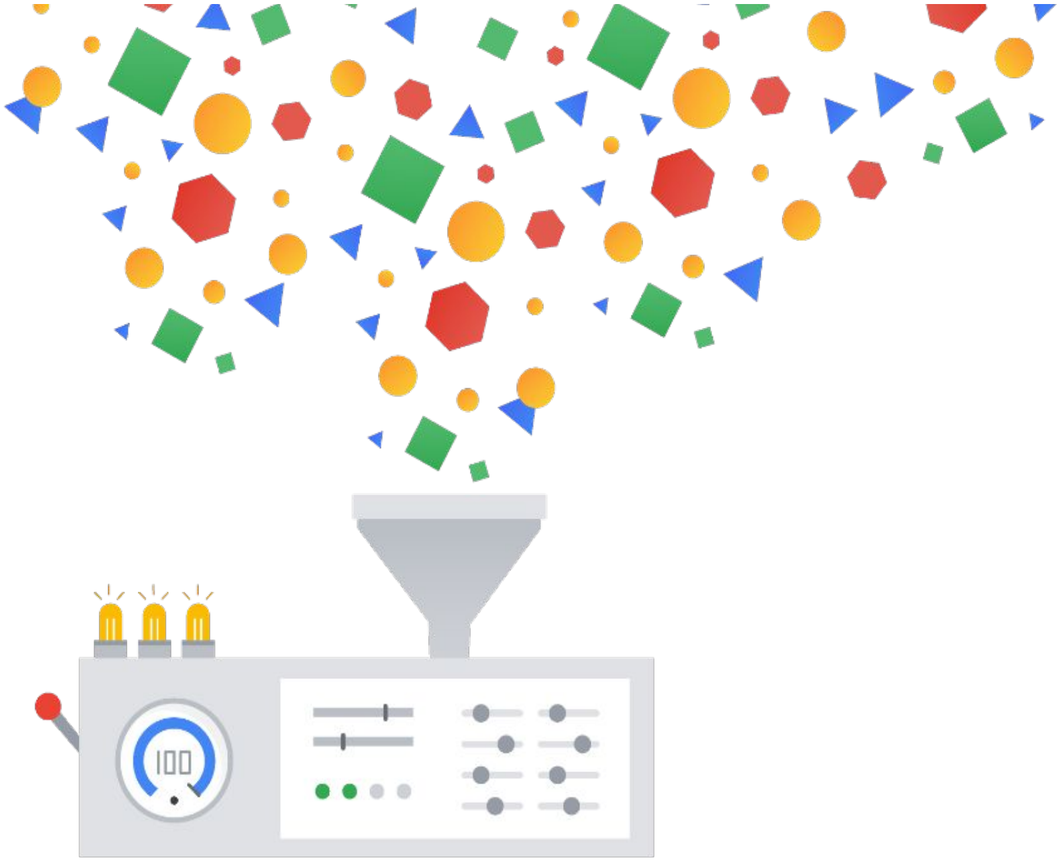
Model

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Pre-trained APIs

Pre-trained APIs don’t require training data



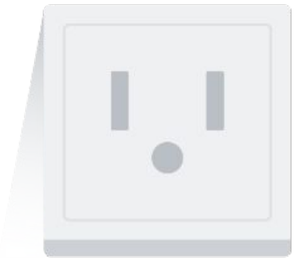
Pre-trained APIs

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Application Programming Interface (API)



Type A



Type B



Type F

- Focus on the API and its required parameters.
- Ignore the underlying implementation.
- Treat APIs like predefined functions.

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Google AI for Python SDK ▸ **Gemini API**

```
import google.generativeai as genai
```

01 Authenticate session.

```
# Configure the API with your key
genai.configure(api_key="YOUR_API_KEY")
```

02 Specify model.

```
# Create a Gemini model instance
model = genai.GenerativeModel('gemini-2.5-flash')
```

03 Make API call.

```
# Send a prompt and get the response
response = model.generate_content("What are the three
largest countries by area?")
```

04 Receive response.

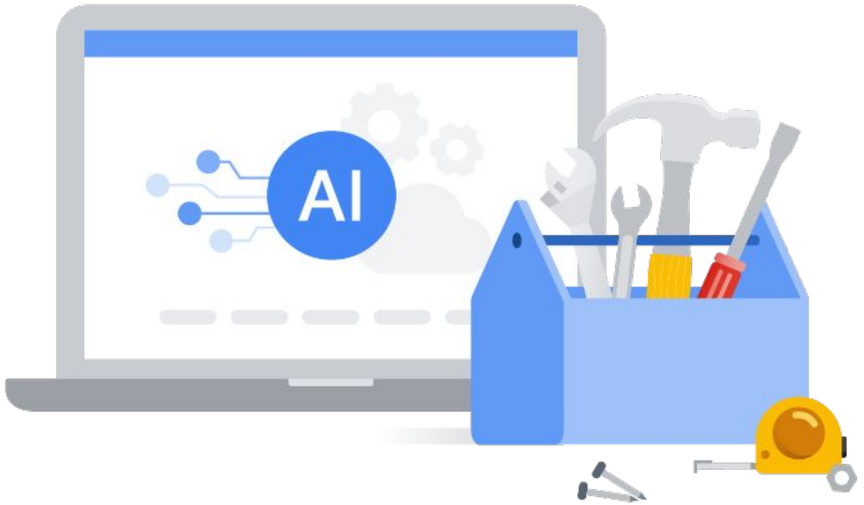
```
# Print the response text
print(response.text)
```

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Custom training

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No-code | No-to-low-code | Low-code | **Code-based**

Gemini Enterprise app
Gemini Enterprise for Customer Experience

Agent Platform
AutoML

Pretrained APIs

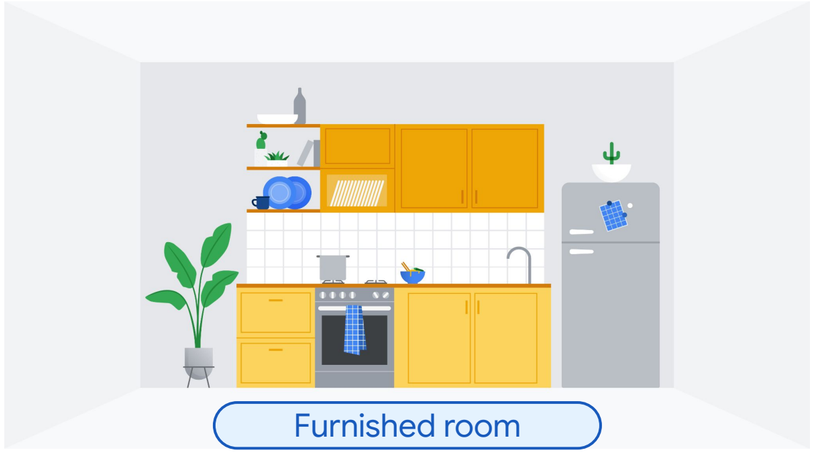
Custom training

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Custom training: Two options

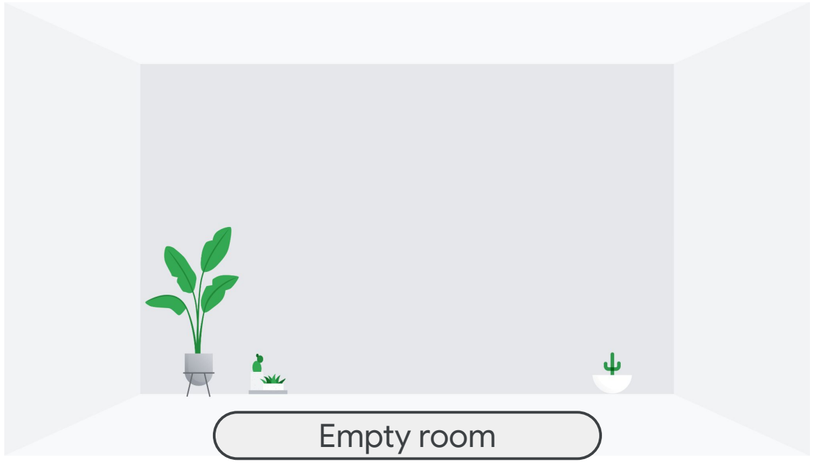
Pre-built container: Pre-installed environment

- Start with a “furnished kitchen”.
- Use common platforms like Python, TensorFlow, and PyTorch.



Custom container: Define your own environment

- Start with an “empty room”.
- Define the environment, machine type, and disks.



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Coding tools with Agent Platform



Agent Platform
Workbench



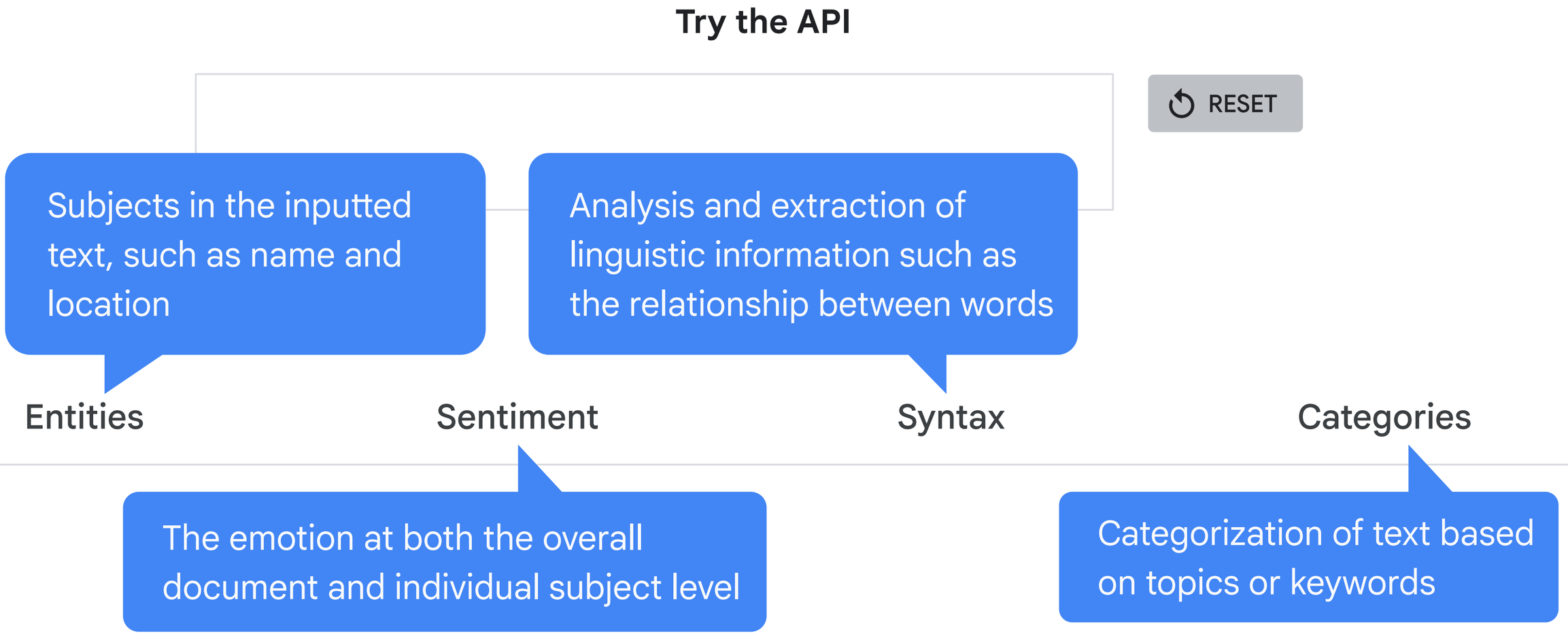
Agent Platform
Colab Enterprise

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DEMO

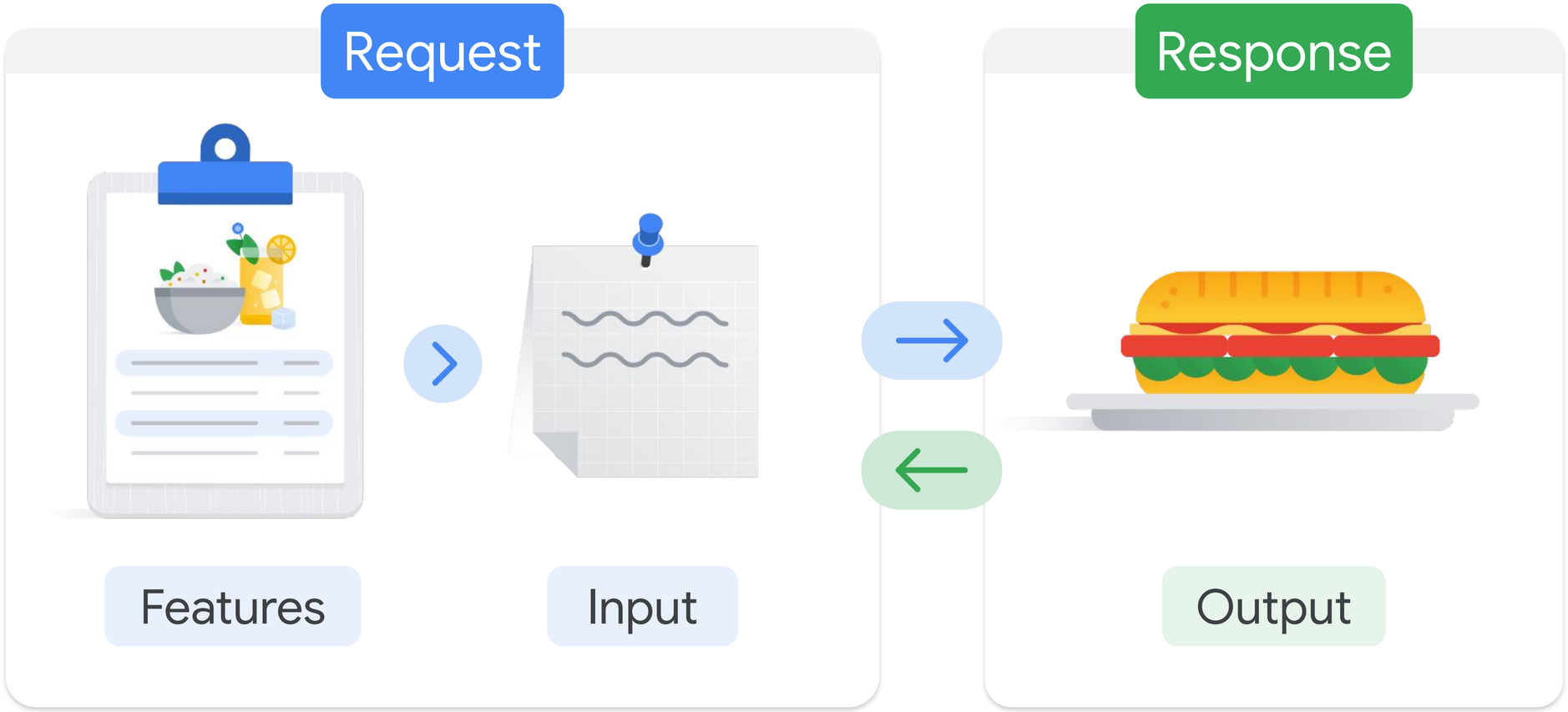
Natural Language API demo

Try the API



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Use APIs in code



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Types of requests

Entities	Sentiment	Syntax	Categories
`analyzeEntities`	`analyzeSentiment` `analyzeEntitySentiment`	`analyzeSyntax`	`classifyText`

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Construct request: request.json

```
{
   "document":{
   "type":"PLAIN_TEXT", 1
   "language":"EN", 2
   "content":" 'Lawrence of Arabia' is a highly rated film biography about British
   Lieutenent T.E. Lawrence. Peter O'Toole plays Lawrence in the film." 3
   },
   "encodingType":"UTF8" 4
}
```

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Call the APIs

```
curl "https://language.googleapis.com/v1/documents:analyzeEntities?key=${API_KEY}" \
  -s -X POST -H "Content-Type: application/json" --data-binary @request.json >
result.json
```

1. curl
2. analyzeEntities
3. @request.json
4. result.json

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Receive responses

Review the result. — or — Parse the result for further usage.

```
cat result.json
```